

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electrical and Electronics Engineering
Academic Year: 2020 – 2021 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE

Course Code & Title: EE8403 Measurements and Instrumentation

Name of the Faculty member: S. Sharmila Kumari

ACTIVITY

Topic: Errors in Measurement

Goal (Learning Outcome):

- ❖ The students will be able to explain various types of errors present in measurements -
(To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: One minute paper

Justification for Choosing the Activity for the topic:

1. Explain various types of errors

The topic Errors in Measurements has 9 types of errors in instruments. Since each type has its own definition, limitation and examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of errors which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 4 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of errors in measurements concept for 2 minutes.
- ✓ Then I told them to write as much as they can within a short period of time (1 minute)
- ✓ Finally, I collected the papers from each column (1 minute)



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students will be able to explain various types of errors present in measurements.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 50% of students answered.**

Reflective Critique:

1. Pre-implementation Reflection :

- **Benefits:**
 - Students can able to attend the question even in the questions are in indirect form.

- Students can able to explain the concepts in examination without any confusion.
- **Challenges:**
 - In the class mostly boys hesitate to answer the questions.
 - Time utilization for conducting activity.
- 2. Post-implementation Reflection :**
- **Benefits:**
 - Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.
- **Challenges:**
 - Slow learners were not able to understand some topics during discussion hours.

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO 12
C212.1	3	3	1	-	1	-	-	1	-	1	-	1

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. H.S. Kalsi, 'Electronic Instrumentation', McGraw Hill, III Edition 2010.

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ACTIVITY

Topic: Principle and types of analog meter

Goal (Learning Outcome):

- ❖ The students will be able to explain the principle and types of analog meter - (To achieve this learning outcome the activity was chosen)

Use of appropriate methods:

Activity Chosen: Demonstration

Justification for Choosing the Activity for the topic:

1. Explain the principle and types of analog meter
The topic Principle and types of analog meter has 7 types of instruments. Since each type has its own principle of operation, torque equation and construction. After teaching the concept, I thought of conducting this activity for making the students to give the difference between the each type of meters which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

Effective presentation:

Details of the Implementation:

After teaching the concept, give students one or two minutes to think about the topic without writing anything.

Total Strength is 29,

Photographer: one student - Mr. V R Ugesh (interested in photography)

Reporter: Myself

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about various types of meters in measurements concept for 2 minutes.
- ✓ Then I told them to Pair with their neighbours and discuss about the construction parts of instruments for another 1 minute.

- ✓ Finally, I show the PMMC and PMMI ammeter for each row (3 minutes)



Significance of Results:

Assessment of Effectiveness/Success of the Activity:

- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
- ✓ While conducting the activity, I understood that the students can able to explain the Analog meters (MC, MI, and Electrodynamometer) construction and working principle.
- ✓ The success of the activity was evaluated by asking the same question in **Internal Assessment test I – Around 80% of students answered.**

Reflective Critique:

3. Pre-implementation Reflection :

- **Benefits:**

- Students can able to attend the question even in the questions are in indirect form.
- Students can able to explain the concepts in examination without any confusion.

- **Challenges:**

- In the class mostly girls hesitate to answer the questions.
- Time utilization for conducting activity.

4. Post-implementation Reflection :

- **Benefits:**

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

- **Challenges:**

- Slow learners were not able to understand some topics during discussion hours.

CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PO1 2
C212.1	3	3	1	-	1	-	-	1	-	1	-	1

References:

1. A.K. Sawhney, 'A Course in Electrical & Electronic Measurements & Instrumentation', Dhanpat Rai and Co, 2010.
2. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/172-demonstrations>